Postcards Analysis Q4

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```r
library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(stringr)

dat <- read_excel("respondents with date.xlsx")
```

```r
dat_long <- dat %>%
  rename(
    id = `New ID Number`,
    group_a = `group a`,
    group_b = `geoup b`,
    group_c = `group c`,
    end_date = `End Date`
  ) %>%
  mutate(
    end_date = as.Date(end_date)
  ) %>%
  pivot_longer(
    cols = c(group_a, group_b, group_c),
    names_to = "group_col",
    values_to = "group"
  ) %>%
  filter(!is.na(group), group != "") %>%
  mutate(
    group = str_trim(group)
  )
names(dat)
```

```
##  [1] "New ID Number"
##  [2] "group a"
##  [3] "geoup b"
##  [4] "group c"
##  [5] "Start Date"
##  [6] "End Date"
##  [7] "...7"
##  [8] "Pre1_Window = Mar 6 - Mar 12          "
##  [9] "Post1_Window = Mar 14 - Mar 20    "
## [10] "...10"
## [11] "Pre2_Window = Apr 3 - Apr 9"
## [12] "Post2_Window = Apr 11 - Apr 17      "
```

Calculation

```r
# ==========================================================
# Build analysis groups
# ----------------------------------------------------------
# GROUP A      = no postcard
# GROUP B      = first postcard recipients (includes C)
# GROUP C      = second postcard recipients
# GROUP B_only = B but not C
# ==========================================================
dat2 <- dat %>%
  rename(
    id = `New ID Number`,
    group_a = `group a`,
    group_b = `geoup b`,
    group_c = `group c`,
    end_date = `End Date`
  ) %>%
  mutate(
    end_date = as.Date(end_date),
    group_a = str_trim(as.character(group_a)),
    group_b = str_trim(as.character(group_b)),
    group_c = str_trim(as.character(group_c))
  )

# Check group counts
dat_groups <- dat2 %>%
  mutate(
    in_A = !is.na(group_a) & group_a != "",
    in_B = !is.na(group_b) & group_b != "",
    in_C = !is.na(group_c) & group_c != "",
    in_B_only = in_B & !in_C
  ) %>%
  filter(!is.na(end_date))

# quick checks
table(dat_groups$in_A)
```

```
##
## FALSE  TRUE
##   474   937
```

```r
table(dat_groups$in_B)
```

```
##
## FALSE  TRUE
##   937   474
```

```r
table(dat_groups$in_C)
```

```
##
```

```
## FALSE  TRUE
##  1259   152
```

```r
table(dat_groups$in_B_only)
```

```
##
## FALSE  TRUE
##  1089   322
```

```r
# ==========================================================
# Create long dataset for daily counts
# ----------------------------------------------------------
# This preserves:
#   GROUP A
#   GROUP B
#   GROUP C
#   GROUP B_only
# Note:
#   GROUP B includes GROUP C by design
# ==========================================================
daily_long <- bind_rows(
  dat_groups %>%
    filter(in_A) %>%
    transmute(id, end_date, group = "GROUP A"),

  dat_groups %>%
    filter(in_B) %>%
    transmute(id, end_date, group = "GROUP B"),

  dat_groups %>%
    filter(in_C) %>%
    transmute(id, end_date, group = "GROUP C"),

  dat_groups %>%
    filter(in_B_only) %>%
    transmute(id, end_date, group = "GROUP B_only")
)

# ==========================================================
# Daily counts
# ==========================================================
daily_counts <- daily_long %>%
  group_by(group, end_date) %>%
  summarise(daily_responses = n(), .groups = "drop") %>%
  complete(
    group = c("GROUP A", "GROUP B", "GROUP C", "GROUP B_only"),
    end_date = seq(min(dat_groups$end_date), max(dat_groups$end_date), by = "day"),
    fill = list(daily_responses = 0)
  )

print(daily_counts)
```

```
## # A tibble: 324 x 3
```

```
##    group   end_date   daily_responses
##    <chr>   <date>               <int>
##  1 GROUP A 2025-02-27             432
##  2 GROUP A 2025-02-28              59
##  3 GROUP A 2025-03-01              38
##  4 GROUP A 2025-03-02              25
##  5 GROUP A 2025-03-03              24
##  6 GROUP A 2025-03-04               6
##  7 GROUP A 2025-03-05               2
##  8 GROUP A 2025-03-06               4
##  9 GROUP A 2025-03-07               1
## 10 GROUP A 2025-03-08               3
## # i 314 more rows
```

```r
# ==========================================================
# Define windows (all windows do not include email reminders)
# ==========================================================
# First postcard sent: Mar 13, 2025
pre1_start  <- as.Date("2025-03-07")
pre1_end    <- as.Date("2025-03-13")
post1_start <- as.Date("2025-03-14")
post1_end   <- as.Date("2025-03-20")

# Second postcard sent: Apr 10, 2025
pre2_start  <- as.Date("2025-04-04")
pre2_end    <- as.Date("2025-04-10")
post2_start <- as.Date("2025-04-11")
post2_end   <- as.Date("2025-04-17")
```

```r
# ==========================================================
# Helper function
# ==========================================================
get_mean_daily <- function(data, group_name, start_date, end_date_window) {
  data %>%
    filter(
      group == group_name,
      end_date >= start_date,
      end_date <= end_date_window
    ) %>%
    summarise(mean_daily = mean(daily_responses, na.rm = TRUE)) %>%
    pull(mean_daily)
}
```

first postcards: March 13

```r
# ==========================================================
#  Postcard 1 analysis
# ----------------------------------------------------------
# Compare GROUP A vs GROUP B
# GROUP B includes everyone who received the first postcard
# ==========================================================
MeanBefore1_A <- get_mean_daily(daily_counts, "GROUP A", pre1_start, pre1_end)
```

```r
MeanAfter1_A  <- get_mean_daily(daily_counts, "GROUP A", post1_start, post1_end)

MeanBefore1_B <- get_mean_daily(daily_counts, "GROUP B", pre1_start, pre1_end)
MeanAfter1_B  <- get_mean_daily(daily_counts, "GROUP B", post1_start, post1_end)

Boost1_A <- MeanAfter1_A - MeanBefore1_A
Boost1_B <- MeanAfter1_B - MeanBefore1_B

DiD1 <- Boost1_B - Boost1_A
```

second postcards: April 10

```r
# ==========================================================
# Postcard 2 analysis
# ----------------------------------------------------------
# Compare GROUP B_only vs GROUP C
# GROUP B_only = first postcard only
# GROUP C      = first + second postcard
# ==========================================================
MeanBefore2_Bonly <- get_mean_daily(daily_counts, "GROUP B_only", pre2_start, pre2_end)
MeanAfter2_Bonly  <- get_mean_daily(daily_counts, "GROUP B_only", post2_start, post2_end)

MeanBefore2_C <- get_mean_daily(daily_counts, "GROUP C", pre2_start, pre2_end)
MeanAfter2_C  <- get_mean_daily(daily_counts, "GROUP C", post2_start, post2_end)

Boost2_Bonly <- MeanAfter2_Bonly - MeanBefore2_Bonly
Boost2_C     <- MeanAfter2_C - MeanBefore2_C

DiD2 <- Boost2_C - Boost2_Bonly
```

```r
# ==========================================================
# Boost tables
# ==========================================================
boost_table_post1 <- tibble(
  postcard = "Postcard 1",
  group = c("GROUP A", "GROUP B"),
  mean_before = c(MeanBefore1_A, MeanBefore1_B),
  mean_after  = c(MeanAfter1_A, MeanAfter1_B),
  boost       = c(Boost1_A, Boost1_B)
)

boost_table_post2 <- tibble(
  postcard = "Postcard 2",
  group = c("GROUP B_only", "GROUP C"),
  mean_before = c(MeanBefore2_Bonly, MeanBefore2_C),
  mean_after  = c(MeanAfter2_Bonly, MeanAfter2_C),
  boost       = c(Boost2_Bonly, Boost2_C)
)

boost_table_all <- bind_rows(boost_table_post1, boost_table_post2)

print(boost_table_post1)
```

```
## # A tibble: 2 x 5
##   postcard   group   mean_before mean_after  boost
##   <chr>      <chr>         <dbl>      <dbl>  <dbl>
## 1 Postcard 1 GROUP A        1.86      0.143 -1.71
## 2 Postcard 1 GROUP B        1.43      1     -0.429
```

```r
print(boost_table_post2)
```

```
## # A tibble: 2 x 5
##   postcard   group        mean_before mean_after  boost
##   <chr>      <chr>              <dbl>      <dbl>  <dbl>
## 1 Postcard 2 GROUP B_only       0.286      0.143 -0.143
## 2 Postcard 2 GROUP C            0          0.571  0.571
```

```r
print(boost_table_all)
```

```
## # A tibble: 4 x 5
##   postcard   group        mean_before mean_after  boost
##   <chr>      <chr>              <dbl>      <dbl>  <dbl>
## 1 Postcard 1 GROUP A            1.86       0.143 -1.71
## 2 Postcard 1 GROUP B            1.43       1     -0.429
## 3 Postcard 2 GROUP B_only       0.286      0.143 -0.143
## 4 Postcard 2 GROUP C            0          0.571  0.571
```

```r
# ==========================================================
# Difference-in-differences table
# ==========================================================
did_table <- tibble(
  postcard = c("Postcard 1", "Postcard 2"),
  treatment_group = c("GROUP B", "GROUP C"),
  control_group   = c("GROUP A", "GROUP B_only"),
  treatment_boost = c(Boost1_B, Boost2_C),
  control_boost   = c(Boost1_A, Boost2_Bonly),
  did_effect      = c(DiD1, DiD2)
)

print(did_table)
```

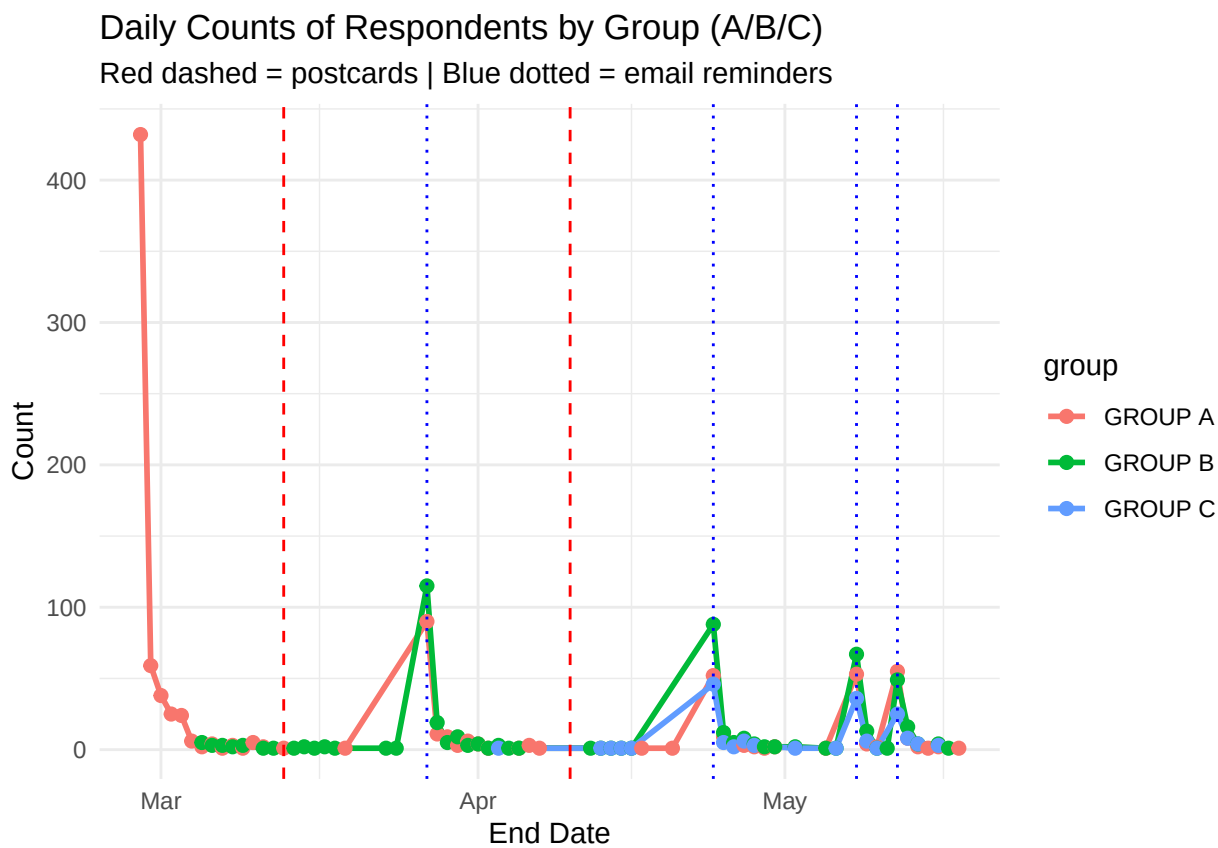
```
## # A tibble: 2 x 6
##   postcard   treatment_group control_group treatment_boost control_boost
##   <chr>      <chr>           <chr>                  <dbl>         <dbl>
## 1 Postcard 1 GROUP B         GROUP A               -0.429        -1.71
## 2 Postcard 2 GROUP C         GROUP B_only           0.571        -0.143
## # i 1 more variable: did_effect <dbl>
```

For the first postcard mailing, the average daily response count declined in both groups, but the decline was substantially smaller among respondents who received the postcard. The difference-in-differences estimate indicates that the postcard increased daily responses by approximately 1.29 responses per day relative to the control group.

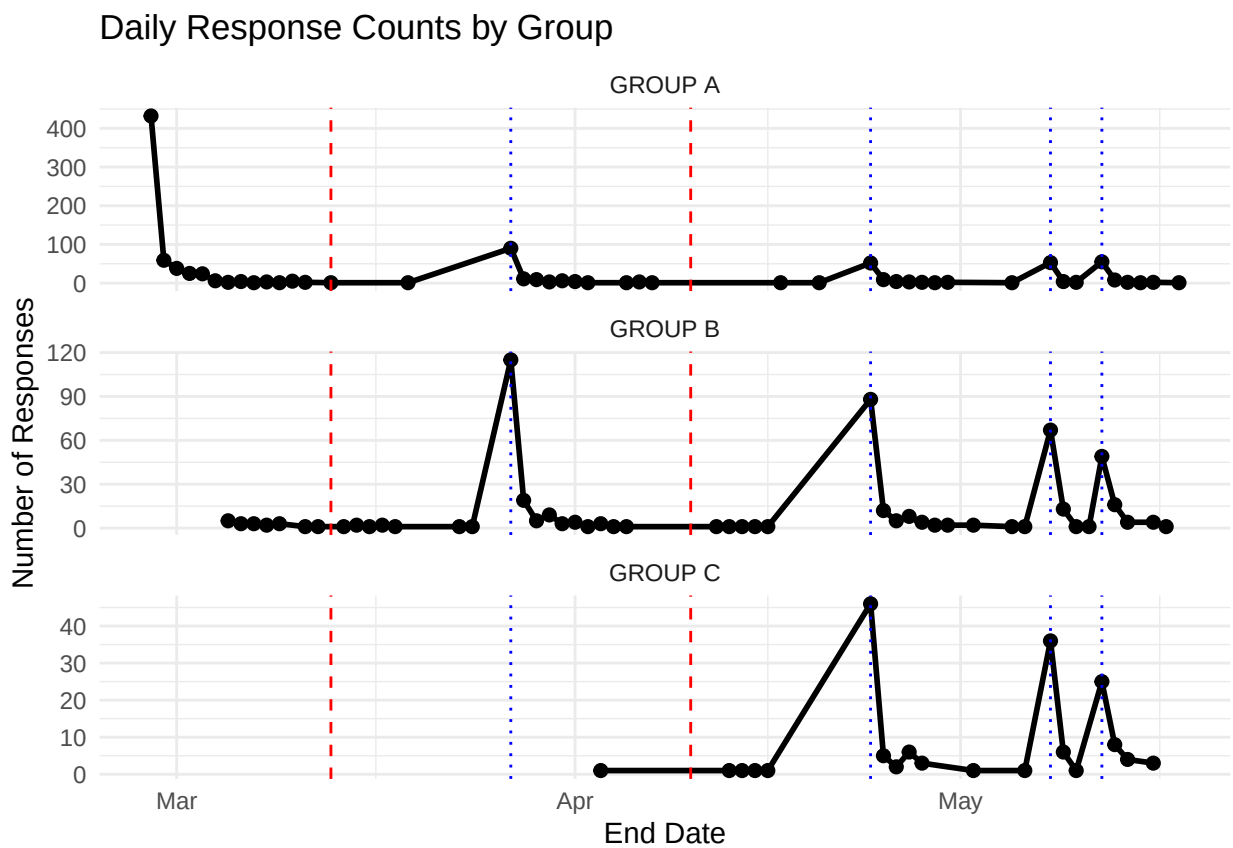
For the second postcard mailing, the group receiving the second postcard exhibited an increase in average daily responses, while the comparison group showed a slight decline. The resulting difference-in-differences estimate suggests an increase of approximately 0.71 responses per day attributable to the second postcard reminder.

plots

```
count_by_date <- dat_long %>%
  group_by(end_date, group) %>%
  summarise(n = n(), .groups = "drop")

ggplot(count_by_date, aes(x = end_date, y = n, color = group)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +

  geom_vline(xintercept = as.Date("2025-03-13"),
             color="red", linetype="dashed", linewidth=0.5) +

  geom_vline(xintercept = as.Date("2025-04-10"),
             color="red", linetype="dashed", linewidth=0.5) +

  geom_vline(xintercept = as.Date("2025-03-27"),
             color="blue", linetype="dotted", linewidth=0.5) +

  geom_vline(xintercept = as.Date("2025-04-24"),
             color="blue", linetype="dotted", linewidth=0.5) +

  geom_vline(xintercept = as.Date("2025-05-08"),
             color="blue", linetype="dotted", linewidth=0.5) +

  geom_vline(xintercept = as.Date("2025-05-12"),
             color="blue", linetype="dotted", linewidth=0.5) +

  labs(
    title="Daily Counts of Respondents by Group (A/B/C)",
    subtitle="Red dashed = postcards | Blue dotted = email reminders",
    x="End Date",
    y="Count"
  ) +
  theme_minimal()
```

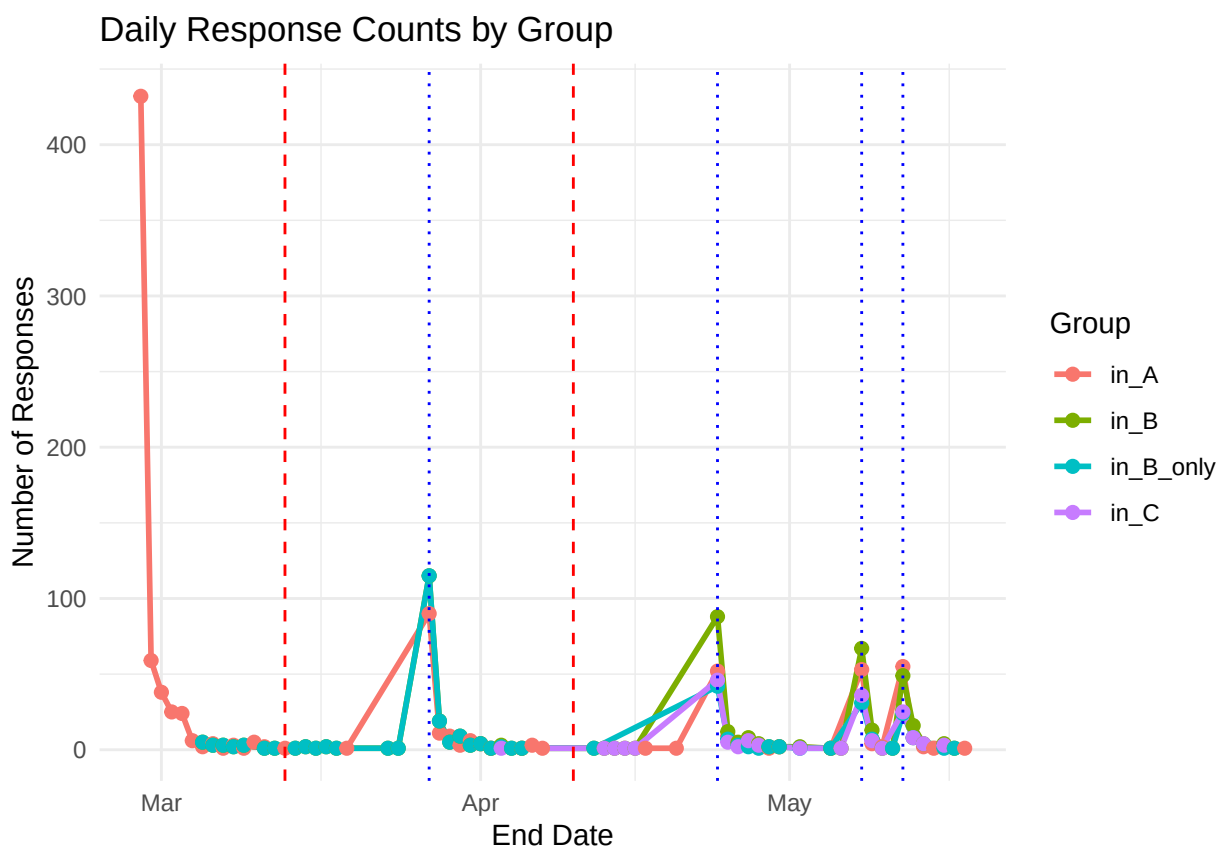


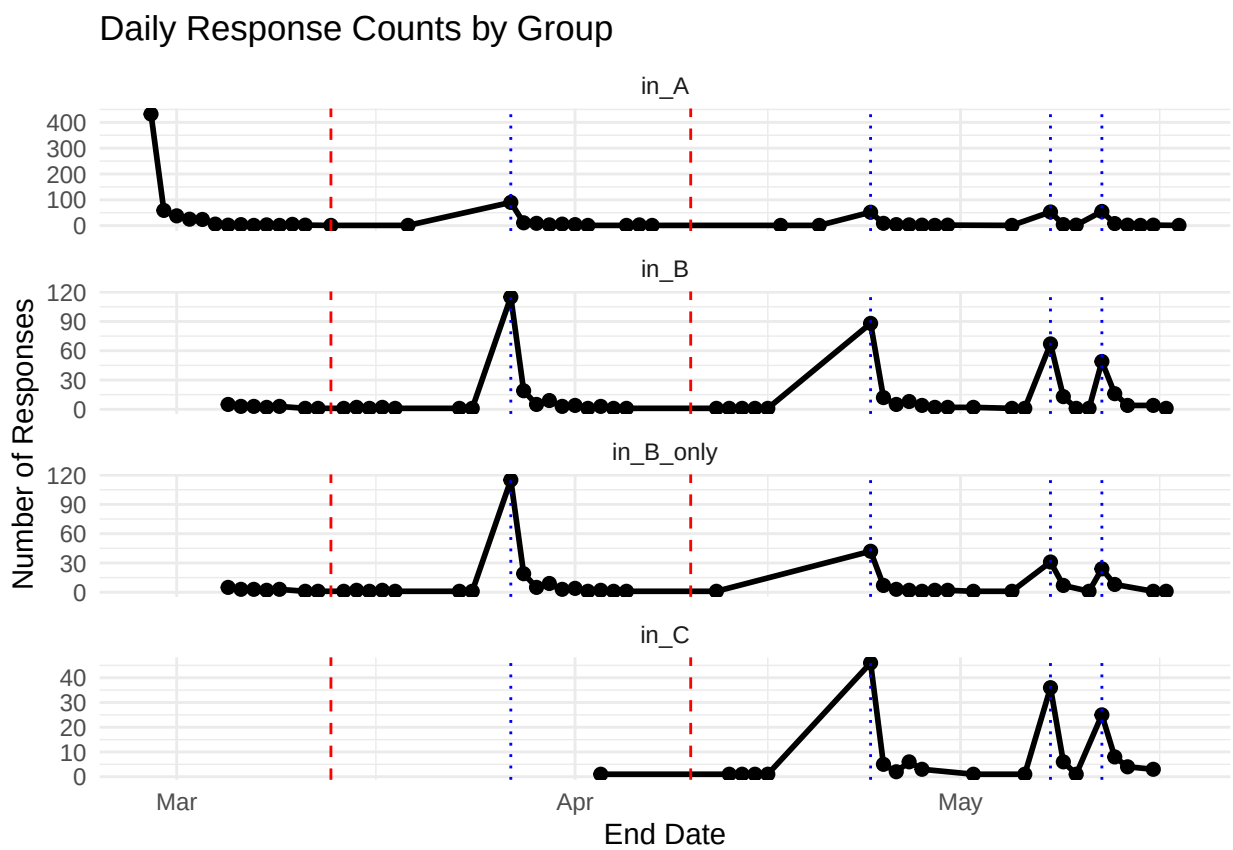

```r
ggplot(count_by_date, aes(x = end_date, y = n)) +
  geom_line(linewidth = 1, color = "black") +
  geom_point(size = 2, color = "black") +

  geom_vline(xintercept = as.Date("2025-03-13"),
             color = "red", linetype = "dashed", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-04-10"),
             color = "red", linetype = "dashed", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-03-27"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-04-24"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-05-08"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-05-12"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +

  facet_wrap(~ group, ncol = 1, scales = "free_y") +

  labs(
    x = "End Date",
    y = "Number of Responses",
    title = "Daily Response Counts by Group"
  ) +
  theme_minimal()
```




```r
daily_long2 <- dat_groups %>%
  pivot_longer(
    cols = c(in_A, in_B, in_C, in_B_only),
    names_to = "group",
    values_to = "in_group"
  ) %>%
  filter(in_group == TRUE) %>%
  group_by(end_date, group) %>%
  summarise(n = n(), .groups = "drop")

ggplot(daily_long2, aes(x = end_date, y = n, color = group)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +

  geom_vline(xintercept = as.Date("2025-03-13"),
             color = "red", linetype = "dashed", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-04-10"),
             color = "red", linetype = "dashed", linewidth = 0.5) +

  geom_vline(xintercept = as.Date("2025-03-27"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-04-24"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-05-08"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-05-12"),
```

```
            color = "blue", linetype = "dotted", linewidth = 0.5) +

  labs(
    x = "End Date",
    y = "Number of Responses",
    color = "Group",
    title = "Daily Response Counts by Group"
  ) +
  theme_minimal()
```




```
ggplot(daily_long2, aes(x = end_date, y = n)) +
  geom_line(linewidth = 1, color = "black") +
  geom_point(size = 2, color = "black") +

  geom_vline(xintercept = as.Date("2025-03-13"),
             color = "red", linetype = "dashed", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-04-10"),
             color = "red", linetype = "dashed", linewidth = 0.5) +

  geom_vline(xintercept = as.Date("2025-03-27"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-04-24"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
  geom_vline(xintercept = as.Date("2025-05-08"),
             color = "blue", linetype = "dotted", linewidth = 0.5) +
```

```r
geom_vline(xintercept = as.Date("2025-05-12"),
           color = "blue", linetype = "dotted", linewidth = 0.5) +

facet_wrap(~ group, ncol = 1, scales = "free_y") +

labs(
  x = "End Date",
  y = "Number of Responses",
  title = "Daily Response Counts by Group"
) +
theme_minimal()
```